Block practice · More blood, heart and circulation

Work through each question type in order. Each section has a worked-example retry and all further practice. Attempt only the part outlined in red.

Explain chamber wall thickness and heart valves by their function

Explain how beta-blockers and ACE inhibitors lower heart rate or pressure

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Practice workbook · Explain chamber wall thickness and heart valves by their function

Why chambers have different muscle thickness and how valves keep flow one-way.

6 question parts: 1 worked-example retry and 5 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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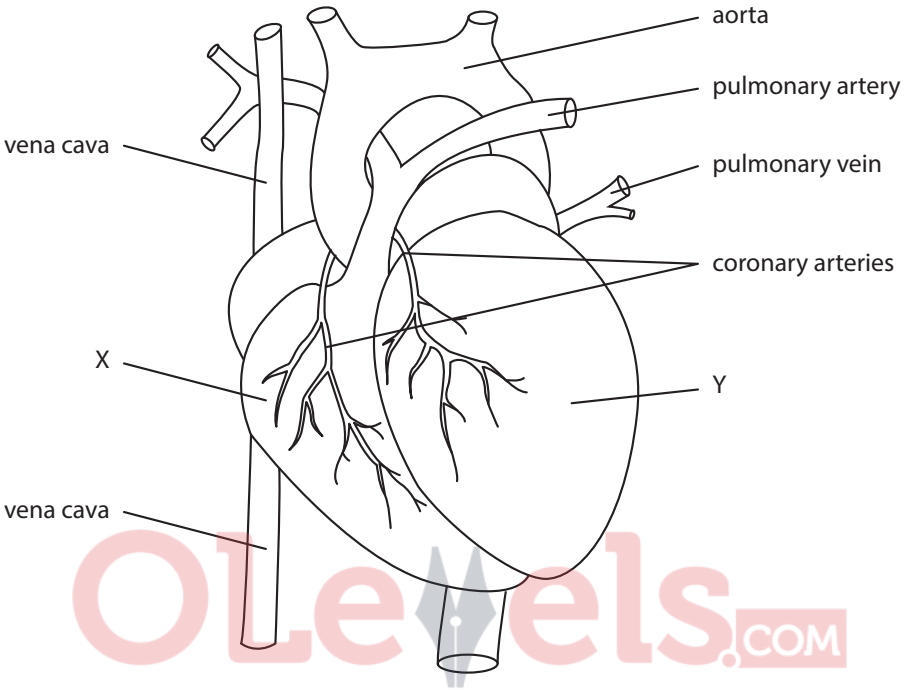
Type ID: HBI-EXPLAIN-HEART-WALLS-AND-VALVES

1. Question 7 · part b · 5 marks

November 2023 Paper 1

Worked-example retry · Attempt part b only.

7 The diagram shows the heart as seen from the front.



(a) The vena cava carries deoxygenated blood.

Name a different blood vessel, shown on the diagram, that also carries deoxygenated blood.

(1)

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1. Question 7 · part b · 5 marks

November 2023 Paper 1

Worked-example retry · Attempt part b only.

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(b) Blood chambers X and Y have walls that are different in thickness.

Explain this difference.

(5)

(c) The coronary arteries can become blocked.

Explain the effect of this blockage on the functioning of the heart.

(3)

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1. Question 7 · part b · 5 marks · Working and answer

Working and answer

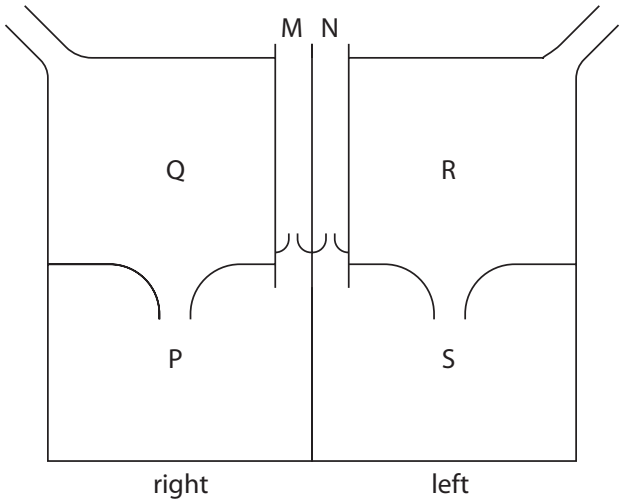
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2. Question 1 · part b.i · 3 marks

January 2022 Paper 1

Further practice · Attempt part b.i only.

1 The diagram shows a human heart in simple form.



(a) (i) Which of these is the name of blood vessel M? **(1)**

- ☒ **A** aorta
- ☒ **B** pulmonary artery
- ☒ **C** pulmonary vein
- ☒ **D** vena cava

(ii) Which of these is the name of blood vessel N? **(1)**

- ☒ **A** aorta
- ☒ **B** pulmonary artery
- ☒ **C** pulmonary vein
- ☒ **D** vena cava

(iii) Draw arrows on the diagram to show the flow of blood through the heart. **(2)**

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2. Question 1 · part b.i · 3 marks

January 2022 Paper 1

Further practice · Attempt part b.i only.

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(iv) Explain why the blood can only flow in the direction of the arrows.

(2)

(b) The width of the walls of each of the four chambers is measured.

These are the results.

0.2 cm 1.0 cm 1.5 cm 0.2 cm

(i) Complete the table by putting the correct width next to the letter of each chamber.

(3)

Chamber	Width in cm
P	
Q	
R	
S	

(ii) Explain the differences in the widths.

(3)

(Total for Question 1 = 12 marks)

2. Question 1 · part b.i · 3 marks · Working and answer

Working and answer

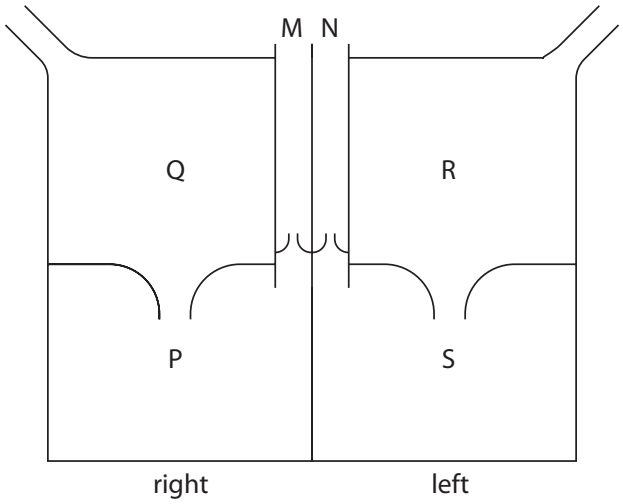
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3. Question 1 · part b.ii · 3 marks

January 2022 Paper 1

Further practice · Attempt part b.ii only.

1 The diagram shows a human heart in simple form.



(a) (i) Which of these is the name of blood vessel M? **(1)**

- ☒ **A** aorta
- ☒ **B** pulmonary artery
- ☒ **C** pulmonary vein
- ☒ **D** vena cava

(ii) Which of these is the name of blood vessel N? **(1)**

- ☒ **A** aorta
- ☒ **B** pulmonary artery
- ☒ **C** pulmonary vein
- ☒ **D** vena cava

(iii) Draw arrows on the diagram to show the flow of blood through the heart. **(2)**

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3. Question 1 · part b.ii · 3 marks

January 2022 Paper 1

Further practice · Attempt part b.ii only.

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(iv) Explain why the blood can only flow in the direction of the arrows.

(2)

(b) The width of the walls of each of the four chambers is measured.

These are the results.

0.2 cm 1.0 cm 1.5 cm 0.2 cm

(i) Complete the table by putting the correct width next to the letter of each chamber.

(3)

Chamber	Width in cm
P	
Q	
R	
S	

DO NOT WRITE IN THIS AREA

(ii) Explain the differences in the widths.

(3)

(Total for Question 1 = 12 marks)

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3. Question 1 · part b.ii · 3 marks · Working and answer

Working and answer

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4. Question 5 · part d · 2 marks

June 2025 Paper 2

Further practice · Attempt part d only.

5 Read the passage below.

Use the information in the passage and your own knowledge to answer the questions that follow.

Losartan belongs to a group of drugs known as angiotensin-II receptor antagonists. The body produces angiotensin-II, which is a substance that binds to receptors in blood vessels, causing them to narrow. This causes an increase in blood pressure. Losartan prevents angiotensin-II binding to these receptors, lowering blood pressure.

If high blood pressure is left untreated, a decrease in kidney function can result in proteinuria. This is a condition that causes high levels of protein to be found in a person's urine. People with high blood pressure and a resultant thickening of the left ventricle can be treated with Losartan to reduce the risk of stroke.

People with high blood pressure should not take Losartan if their liver is not functioning properly. Women who are pregnant or who are breast feeding should not take Losartan.

(a) Describe how the action of Losartan differs from the action of an ACE inhibitor. (lines 4 and 5)

(4)

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4. Question 5 · part d · 2 marks

June 2025 Paper 2

Further practice · Attempt part d only.

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(b) Explain how the structure of a named type of blood vessel allows angiotensin-II to increase blood pressure. (lines 2 to 5)

(4)

(c) Explain why proteinuria is a sign of problems with kidney function. (lines 6 to 8)

(2)

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(d) Suggest why high blood pressure causes the left ventricle to become thickened. (lines 8 to 9)

(2)

(e) Explain why people who have high blood pressure are advised not to take Losartan if their liver is not functioning properly. (lines 10 to 11)

(2)

(Total for Question 5 = 14 marks)

4. Question 5 · part d · 2 marks · Working and answer

Working and answer

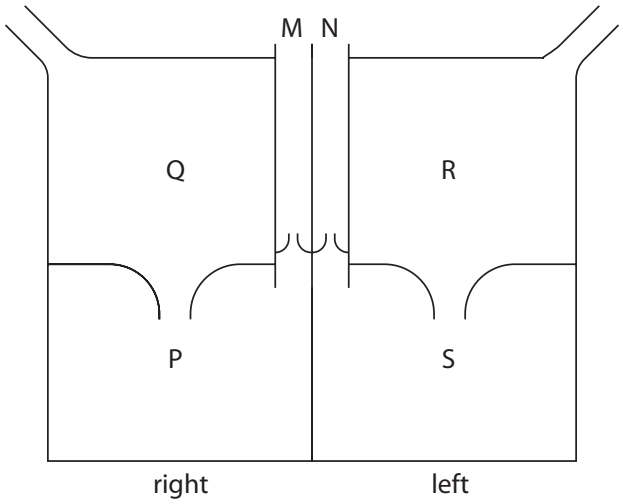
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5. Question 1 · part a.iv · 2 marks

January 2022 Paper 1

Further practice · Attempt part a.iv only.

1 The diagram shows a human heart in simple form.



(a) (i) Which of these is the name of blood vessel M? **(1)**

- ☒ **A** aorta
- ☒ **B** pulmonary artery
- ☒ **C** pulmonary vein
- ☒ **D** vena cava

(ii) Which of these is the name of blood vessel N? **(1)**

- ☒ **A** aorta
- ☒ **B** pulmonary artery
- ☒ **C** pulmonary vein
- ☒ **D** vena cava

(iii) Draw arrows on the diagram to show the flow of blood through the heart. **(2)**

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5. Question 1 · part a.iv · 2 marks

January 2022 Paper 1

Further practice · Attempt part a.iv only.

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(iv) Explain why the blood can only flow in the direction of the arrows.

(2)

(b) The width of the walls of each of the four chambers is measured.

These are the results.

0.2 cm 1.0 cm 1.5 cm 0.2 cm

(i) Complete the table by putting the correct width next to the letter of each chamber.

(3)

Chamber	Width in cm
P	
Q	
R	
S	

(ii) Explain the differences in the widths.

(3)

(Total for Question 1 = 12 marks)

5. Question 1 · part a.iv · 2 marks · Working and answer

Working and answer

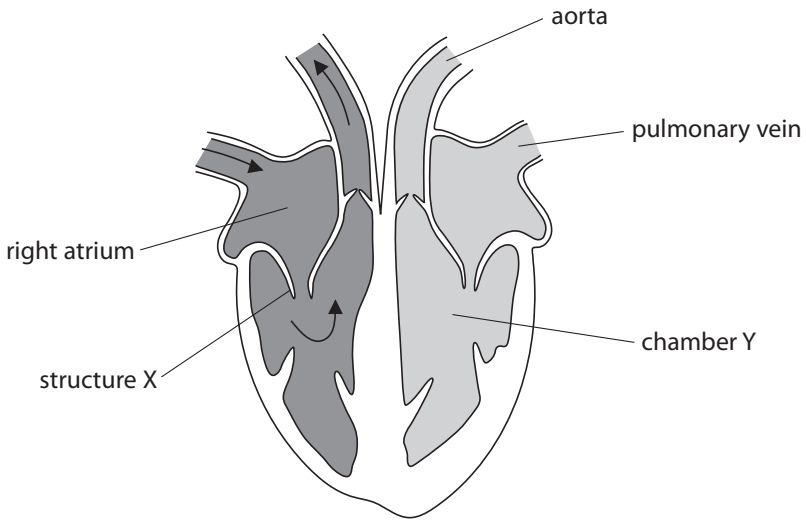
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6. Question 2 · part a.ii · 2 marks

January 2023 Paper 2

Further practice · Attempt part a.ii only.

2 (a) The diagram shows a section through the heart.



(i) Name chamber Y. **(1)**

(ii) Explain the function of structure X. **(2)**

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P 7 2 6 0 2 A 0 4 2 8

6. Question 2 · part a.ii · 2 marks

January 2023 Paper 2

Further practice · Attempt part a.ii only.

(iii) The table lists some properties of blood.

Add ticks (✓) to those boxes that show the properties of blood in the aorta.

(2)

Properties of blood	Blood in the aorta
oxygenated	
deoxygenated	
high pressure	
medium pressure	
low pressure	

(iv) State where blood in the left side of the heart has come from.

(1)

.....

(b) In one year in the United Kingdom, there were 173 heart transplants.

Heart transplant operations require hearts to be donated.

Some hearts cannot be used because the donor is too old.

Give two other reasons why a heart might not be suitable for a transplant.

(2)

1 .....

.....

2 .....

.....

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6. Question 2 · part a.ii · 2 marks · Working and answer

Working and answer

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Practice workbook · Explain how beta-blockers and ACE inhibitors lower heart rate or pressure

Mechanisms and effects of beta-blockers, ACE inhibitors and related receptor blockers.

6 question parts: 1 worked-example retry and 5 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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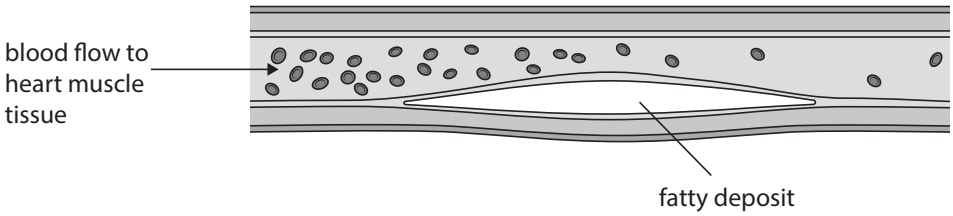
Type ID: HBI-EXPLAIN-CARDIOVASCULAR-DRUG-ACTION

1. Question 7 · part c · 3 marks

June 2024 Paper 2

Worked-example retry · Attempt part c only.

7 (a) The diagram shows a fatty deposit in an artery carrying blood to the heart muscle.



(Source: adapted from https://www.heartandstroke.ca/heart-disease/conditions/atherosclerosis)

Explain how a build-up of a fatty deposit in an artery carrying blood to the heart muscle can cause heart disease.

(4)

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1. Question 7 · part c · 3 marks

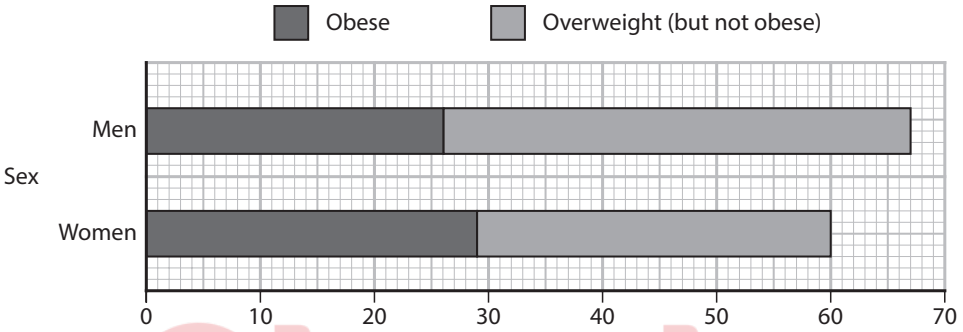
June 2024 Paper 2

Worked-example retry · Attempt part c only.

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(b) People with obesity are more at risk of heart disease than those who are overweight.

The chart shows the percentage of men and women in the United Kingdom who were either overweight or obese in 2018.



(Source: https://digital.nhs.uk/data-and-information/publications/statistical/health-survey-for-england/2018/summary)

(i) In 2018 there were 66.5 million people living in the UK. Of these people, 33 million were men.

Calculate how many more women than men were classified as obese.

(3)

extra number of women = ...

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P 7 4 6 3 0 A 0 1 9 2 4

1. Question 7 · part c · 3 marks

June 2024 Paper 2

Worked-example retry · Attempt part c only.

(ii) Describe a calculation that could be used to determine if a person is obese.

(3)

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20

P 7 4 6 3 0 A 0 2 0 2 4

1. Question 7 · part c · 3 marks

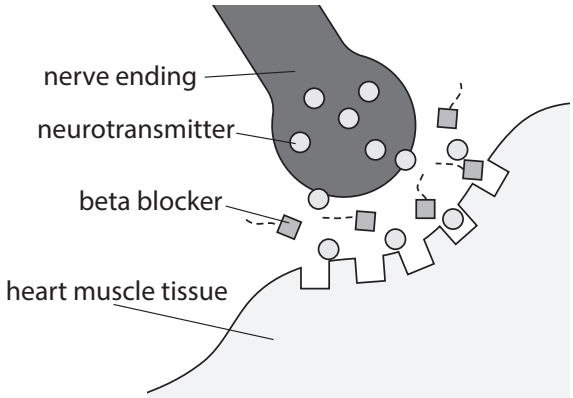
June 2024 Paper 2

Worked-example retry · Attempt part c only.

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(c) The diagram shows a junction between a neurone and heart muscle.



(Source: Adapted from: https://study.com/academy/lesson/what-are-beta-blockers-uses-types-examples-side-effects.html)

Use the diagram to explain how beta blockers can reduce the risk of heart disease.

(3)

(Total for Question 7 = 13 marks)

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1. Question 7 · part c · 3 marks · Working and answer

Working and answer

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2. Question 9 · part a.i · 1 mark

January 2020 Paper 1

Further practice · Attempt part a.i only.

9 In sports such as archery and shooting, beta-blockers are used illegally by some competitors to help reduce the effects of stress and anxiety, which can affect performance.

(a) (i) Suggest why the use of beta-blockers in competitive sport is illegal.

(1)

(ii) Stress and anxiety can cause an increase in the release of adrenaline.

Explain the effects of adrenaline on the body.

(4)

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24

P 5 9 9 0 9 A 0 2 4 2 8

2. Question 9 · part a.i · 1 mark · Working and answer

Working and answer

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3. Question 9 · part b.i · 3 marks

January 2020 Paper 1

Further practice · Attempt part b.i only.

9 In sports such as archery and shooting, beta-blockers are used illegally by some competitors to help reduce the effects of stress and anxiety, which can affect performance.

(a) (i) Suggest why the use of beta-blockers in competitive sport is illegal.

(1)

(ii) Stress and anxiety can cause an increase in the release of adrenaline.

Explain the effects of adrenaline on the body.

(4)

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24

P 5 9 9 0 9 A 0 2 4 2 8

3. Question 9 · part b.i · 3 marks

January 2020 Paper 1

Further practice · Attempt part b.i only.

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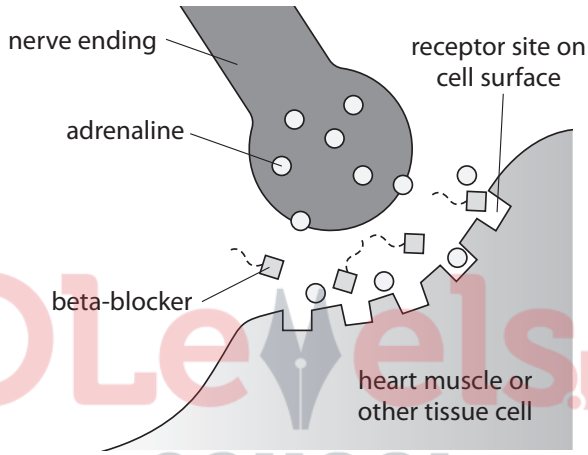
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(b) Adrenaline acts as a hormone and as a neurotransmitter.

Beta-blockers act at the junction between nerve endings and body tissues where adrenaline is found.

The diagram shows the action of beta-blockers at a synapse.



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(i) Explain how beta-blockers reduce symptoms of stress and anxiety.

(3)

(ii) Suggest how the action of beta-blockers would be a disadvantage in active sports such as athletics.

(2)

(Total for Question 9 = 10 marks)

3. Question 9 · part b.i · 3 marks · Working and answer

Working and answer

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4. Question 9 · part b.ii · 2 marks

January 2020 Paper 1

Further practice · Attempt part b.ii only.

9 In sports such as archery and shooting, beta-blockers are used illegally by some competitors to help reduce the effects of stress and anxiety, which can affect performance.

(a) (i) Suggest why the use of beta-blockers in competitive sport is illegal.

(1)

(ii) Stress and anxiety can cause an increase in the release of adrenaline.

Explain the effects of adrenaline on the body.

(4)

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P 5 9 9 0 9 A 0 2 4 2 8

4. Question 9 · part b.ii · 2 marks

January 2020 Paper 1

Further practice · Attempt part b.ii only.

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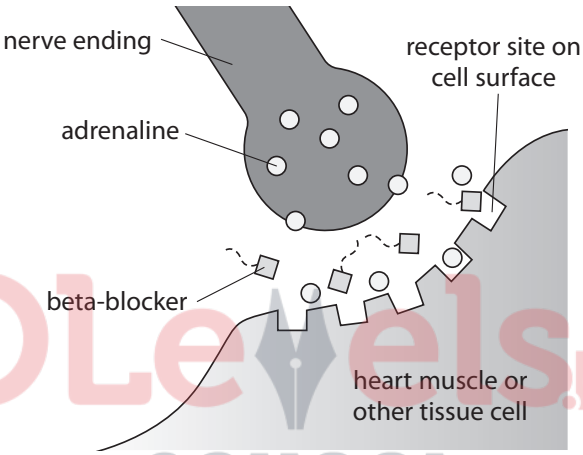
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Beta-blockers act at the junction between nerve endings and body tissues where adrenaline is found.

The diagram shows the action of beta-blockers at a synapse.



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(i) Explain how beta-blockers reduce symptoms of stress and anxiety.

(3)

(ii) Suggest how the action of beta-blockers would be a disadvantage in active sports such as athletics.

(2)

(Total for Question 9 = 10 marks)

4. Question 9 · part b.ii · 2 marks · Working and answer

Working and answer

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5. Question 9 · part a · 4 marks

June 2022 Paper 1

Further practice · Attempt part a only.

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9 In England, 31% of men and 26% of women have high blood pressure.

(a) Explain how medication can be used to treat people with high blood pressure.

(4)

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(b) (i) Describe what is meant by systolic blood pressure.

(2)

(ii) Describe what is meant by diastolic blood pressure.

(2)

(Total for Question 9 = 8 marks)

5. Question 9 · part a · 4 marks · Working and answer

Working and answer

OLevels.com SCHOOL

6. Question 5 · part a · 4 marks

June 2025 Paper 2

Further practice · Attempt part a only.

5 Read the passage below.

Use the information in the passage and your own knowledge to answer the questions that follow.

Losartan belongs to a group of drugs known as angiotensin-II receptor antagonists. The body produces angiotensin-II, which is a substance that binds to receptors in blood vessels, causing them to narrow. This causes an increase in blood pressure. Losartan prevents angiotensin-II binding to these receptors, lowering blood pressure.

If high blood pressure is left untreated, a decrease in kidney function can result in proteinuria. This is a condition that causes high levels of protein to be found in a person's urine. People with high blood pressure and a resultant thickening of the left ventricle can be treated with Losartan to reduce the risk of stroke.

People with high blood pressure should not take Losartan if their liver is not functioning properly. Women who are pregnant or who are breast feeding should not take Losartan.

(a) Describe how the action of Losartan differs from the action of an ACE inhibitor. (lines 4 and 5)

(4)

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6. Question 5 · part a · 4 marks

June 2025 Paper 2

Further practice · Attempt part a only.

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(b) Explain how the structure of a named type of blood vessel allows angiotensin-II to increase blood pressure. (lines 2 to 5)

(4)

(c) Explain why proteinuria is a sign of problems with kidney function. (lines 6 to 8)

(2)

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(d) Suggest why high blood pressure causes the left ventricle to become thickened. (lines 8 to 9)

(2)

(e) Explain why people who have high blood pressure are advised not to take Losartan if their liver is not functioning properly. (lines 10 to 11)

(2)

(Total for Question 5 = 14 marks)

6. Question 5 · part a · 4 marks · Working and answer

Working and answer

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